

Retail Strategy and Analytics – Data Analysis Report

1. Project Overview

This analysis focuses on understanding customer purchasing behavior for chip products using two datasets:

- Customer data (Lifestage & Subscription type)
- Transaction data (Store, product, quantity, sales)

The objective was to:

- Identify customer segments contributing most to chip sales
- Understand brand and pack-size preferences
- Detect anomalies, clean data, and generate actionable insights for retail strategy

2. Data Preparation

2.1 Merging & Cleaning

Key steps included:

- Merging the transactions and customer datasets on loyalty card numbers.
- Formatting dates correctly from Excel numeric date format.
- Removing Salsa products (not categorized as chips).
- Detecting outliers:
 - A customer with card number 226000 purchased 200 units twice—these were removed as anomalies.
- Limiting quantities to transactions ≤ 5 units to remove extreme outliers.
- Cleaning product names, extracting:
 - Brand
 - Pack size

2.2 Missing Date Handling

- Found one missing date: 2018-12-25 (Christmas Day).
- As stores were closed, the missing date was acknowledged but not artificially filled with transactions.

3. Sales Trends

3.1 Daily and Monthly Trends

- Sales peak in December, especially before Christmas (Dec 22–24).
- Sales drop sharply on December 25 because stores were closed.
- Overall monthly fluctuations show:

- Highest sales: December
- Lowest sales: February

3.2 Demand Insight

Consumer demand increases heavily during the holiday season.

4. Product Analysis

4.1 Pack Size

Most common pack sizes:

1. 175g (most popular)
2. 150g
3. 134g

This pattern holds true across all customer segments.

4.2 Brand Popularity

Across all segments:

- Kettle is the most sold brand.
- Followed by:
 - Smiths
 - Doritos
 - Pringles

Segment-specific differences were observed but Kettle dominates consistently.

5. Customer Segmentation Analysis

Segments are defined by:

- Lifestage (Young Singles/Couples, Families, Retirees, etc.)
- Subscription type (Budget, Mainstream, Premium)

5.1 Highest Spending Segments

By total sales:

1. Older Families – Budget
2. Young Singles/Couples – Mainstream
3. Retirees – Mainstream

5.2 Number of Customers per Segment

- Largest population segments:
 - Young Singles / Couples – Mainstream (7917 customers)

- Retirees – Mainstream (6358 customers)
- Older Singles / Couples – Mainstream (4858 customers)

5.3 Average Quantity Bought

Segments buying the most chips per customer:

1. **Older Families – Mainstream**
2. **Older Families – Budget / Premium**
3. **Young Families – Budget / Premium**

This highlights family-oriented categories purchase more quantity per visit.

5.4 Average Unit Price Willingness

Segments paying highest *unit price* per pack:

- **Young Singles/Couples – Mainstream**
- **Midage Singles/Couples – Mainstream**
- **Retirees – Budget/Premium**

These groups are **more willing to pay premium prices**.

6. Brand Preference by Segment

Across nearly all 21 Group–Subscription combinations:

- **Kettle remains the top choice**
- Second choice varies:
 - **Smiths** (dominant for most segments)
 - **Doritos** (especially for Young Singles/Couples – Mainstream)

Key standout:

- **Young Singles/Couples – Mainstream prefer Doritos strongly**, unlike other groups.

7. Pack Size Preference by Segment

Across every segment:

1. **175g packs** were the most purchased.
2. Followed by **150g**.
3. Then **134g**.

This indicates a strong universal preference for mid-sized chip packs.

8. Statistical Test

A t-test comparing:

- Mainstream Young & Midage Singles/Couples vs
 - Budget/Premium equivalents
- showed:

p-value $< 1e^{-237}$, indicating a **significant difference** in average spending per transaction.

Mainstream groups spend more.

9. Key Insights

Customer Behavior Insights

- Families—especially **Older Families**—are the biggest contributors to chip sales.
- **Young Singles/Couples Mainstream** is a large and valuable segment.
- **Retirees** are a stable, high-volume customer base.

Brand Insights

- **Kettle** is the preferred brand for all segments.
- Smiths follows closely.
- Doritos is uniquely preferred by Young Singles/Couples (Mainstream).

Pack Size Insights

- **175g packs dominate** universally.
- Promotions for 175g should appeal to all groups.

Seasonal Insights

- December shows a surge in sales.
 - Opportunity for targeted holiday promotions.
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10. Recommendations

1. Target Older Families – Budget

- Offer **bundle deals** and **bulk purchase promotions**.
- Use loyalty rewards to increase repeat purchases.

2. Focus on Young Singles/Couples – Mainstream

- Promote **Doritos** and **Kettle** via:
 - In-store displays

- Social media campaigns targeting young adults
- Use value-based promotions or multipacks.

3. Leverage Retirees – Mainstream

- Provide prominent catalogue promotions.
- Use traditional advertising channels (printed catalogues, in-store posters).

4. Product Strategy

- Prioritize stock, displays, and promotions for:
 - **175g packs**
 - **Kettle & Smiths brands**

5. Seasonal Strategy

- Increase stock and promotional offers between **Dec 1–24**.
- Create holiday-specific bundles.